

TicTac Tetris – Team Project

ASE 420

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Project Description

TicTac Tetris is our team project to further develop our software design and team building skills to better ourselves for our future careers.

Problem Domain

As it stands our current Tetris game is very bare bones and unpolished, the goal of this project is to improve on our current base game using our software design principles and add new and unique features with our software engineering skills.

Core Features Overview

- UI Overhaul
- Challenges
- Mission Mode
- Quality of Life Updates

Feature 1: UI Overhaul

Owner: Tanner Andrews

Description

Polish the game's UI and make it feel more user friendly

Requirements

- Navigation layout
- Style design (colors, art, other visuals)
- Clear & Visible gameplay

Feature 2: Challenges

Owner: Tanner Andrews

Description

Tetris gameplay with unique effects that change how it is played

Requirements

- Menu screen to select which challenges will be active
- Unique challenges such as new blocks, screen size, or gravity
- Score multipliers for each unique challenge selected

Feature 3: Mission Mode

Owner: Joseph Gallucci

Description

Users will have to clear the board of a pre-determined obstacle to complete their "mission"

Requirements

- Mission selection screen
- Unique missions for players to complete
- Mission replayability

Feature 4: Quality of Life Updates

Owner: Joseph Gallucci

Description

Modern iterations of Tetris have niche features to make the game more enjoyable

Requirements

- Quick dropping of blocks
- "Wall kicks" for base game elements
- High scores saved locally

Total Features and Requirements

Features: 4

Requirements: 12

Architecture

The architecture serves to separate the game into clear components so the code is easier to maintain and expand.

- Reduce global variables
- Give each component a clear responsibility
- Create a foundation for future features

Design

Refactor the existing procedural code into an organized object-oriented structure.

- Encapsulation
- Single Responsibility
- Maintainability
- Extensibility

Team Rules

1. Communicate effectively and consistently.
2. All weekly updates must be finished by Saturday of the current week.
3. Communication will be handled over discord weekly for quick messages and questions.

Schedule & Milestones

Sprint 1

- Project setup
- Initial Analysis of design issues
- Initial implementation of new features
- Initial refactoring of base tetris game

Schedule & Milestones

Sprint 2

- Retrospective of sprint 1
- Completely polished original game
- Completion of expected features
- Possible micro additions if time allows

Project Documentation

- [Project Plan Presentation \(PPP\)](#)
- [GitHub Repository](#)